LAYER 1 · PHYSICAL

L1

Copper, Pins and Hardware

Everything starts with a working link: wiring standards, terminating and testing a cable, how a building is cabled, and a calm order for fixing hardware.

YASIR ABDULLAH · YASIR-PORTFOLIO-ZETA.VERCEL.APP/NOTES/EBOOKS/LAYER-1-PHYSICAL

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CHAPTER 1 · T568A / T568B

T568A and T568B

The two colour orders for an RJ45 plug, and why one cable uses one standard at both ends.



Pin 1 to pin 8, looking at the plug with the clip underneath. Only the orange and green pairs swap.

An Ethernet cable has eight copper wires twisted into four pairs. T568A and T568B are the two agreed orders for putting those wires into the eight pins of an RJ45 plug. They differ in one way only: the orange pair and the green pair swap places.

Reading the pins

Hold the plug with the copper contacts facing you and the clip underneath. Pin 1 is on the left.

PIN	T568A	T568B	10/100 MB/S USE
1	white-green	white-orange	transmit +
2	green	orange	transmit –
3	white-orange	white-green	receive +
4	blue	blue	unused
5	white-blue	white-blue	unused

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PIN	T568A	T568B	10/100 MB/S USE
6	orange	green	receive –
7	white-brown	white-brown	unused
8	brown	brown	unused

Notice pins 3 and 6. The receive pair is split around the blue pair. That is deliberate, and it is the reason you cannot simply lay the colours in pairs side by side.

Gigabit Ethernet and PoE use all four pairs, so "unused" only applies to old 10/100 links.

Straight-through and crossover

- **Straight-through:** the same standard at both ends (B to B, or A to A). This is the cable for almost everything: PC to switch, switch to router, patch panel to switch.
- **Crossover:** A at one end and B at the other. It was needed to join two similar devices, such as switch to switch. Modern ports use **Auto MDI-X** and sort this out themselves, so crossover cables are now rare.

> TIP
>
> Pick one standard for the whole site and write it down. Most installers use T568B. Mixing standards in one building is how a "straight" cable turns into a crossover by accident.

When it goes wrong

- **Link up at 100 Mb/s instead of 1 Gb/s:** one of the blue or brown pairs is broken or badly seated. The link falls back to the two pairs that work.
- **No link at all:** usually pin 1, 2, 3 or 6 is open or swapped.
- **Works, but with errors:** a split pair. Each pin goes to the right pin, but the two wires of a pair were separated, so the twist no longer cancels noise. Only a tester that checks pairs, not just continuity, will catch it.

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Crimping and Testing a Cable

Terminating an RJ45 plug that passes first time, and reading what a cable tester tells you.



The six steps, then a wiremap: the tester lights each pin in turn. Here pin 7 is open.

A badly crimped plug is the cheapest fault in networking to create and one of the slowest to find. Done in the right order, a termination passes the tester first time.

The six steps

1. **Strip** about 25 mm of the outer jacket. Score it gently; a nick in the inner insulation causes faults months later.
2. **Untwist** the pairs only as far as you need. Every millimetre of untwisted wire lets in more crosstalk.
3. **Order** the wires to your standard. For T568B: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.
4. **Trim** the wires straight across to about 13 mm, so they reach the front of the plug with the jacket still inside it.
5. **Insert** the wires with the clip facing down. Look through the end of the plug: each copper tip must touch the front wall.
6. **Crimp** firmly in one squeeze. The strain-relief wedge should grip the outer jacket, not bare wires.

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If any coloured wire is visible outside the back of the plug, cut it off and start again. A plug that only holds the inner wires will work today and fail when someone moves the desk.

Reading the tester

A basic tester has a main unit and a remote. It lights each pin in turn, one to eight, at both ends.

WHAT YOU SEE	WHAT IT MEANS	FIX
1 to 8 light in order	Every pin is connected straight	Done
One number never lights	**Open**: that wire is not connected	Re-terminate the end
Two numbers light at once	**Short**: two wires touch	Re-terminate; check f
Lights out of order	**Crossed** or wrong standard	Compare both ends t

A continuity tester cannot see a **split pair**. A certifier or qualification tester measures crosstalk and length as well, and is the tool to use on permanent cabling.

A habit worth keeping

Test every cable you make before it leaves your hands, and label both ends. A two-second test at the bench saves an hour in the ceiling.

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Structured Cabling

How a building is wired: outlets, horizontal runs, patch panels and the backbone.



A desk reaches the core through fixed, labelled pieces. Only the two short patch cords ever move.

Structured cabling is the idea that a building's network should be wired once, neatly and to a plan, so that moving a desk means moving a patch cord, not pulling a new cable through the ceiling.

The pieces, from desk to core

and the short patch cord to the wall.

ed jack, for example D-214.

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3. **Horizontal cable:** the permanent run inside the wall or ceiling, from the outlet to the telecom room. It is solid-core copper and at most **90 m** long.
4. **Patch panel:** where every horizontal cable ends, one port per outlet, in the same order and with the same labels.
5. **Patch cord:** a short flexible cable from the panel port to a switch port.
6. **Backbone:** fibre or copper between the floor's telecom room (IDF) and the main room (MDF).

The 100 m limit for Ethernet is shared out as 90 m of permanent cable plus up to 10 m of patch cords at both ends together.

Why the patch panel matters

Switches are replaced every few years. The cable in the walls should last twenty. The patch panel separates the two: the fixed cabling never touches the switch, so a failed switch port is fixed by moving one patch cord.

Labels are part of the job

Use one scheme everywhere and put the same label on the outlet, the cable and the panel port.

```text
D-214     floor 2, room 14, outlet D
PP2-14    patch panel 2, port 14
Gi1/0/14  switch port it is patched to
```

Keep a simple sheet that maps outlet to panel port to switch port. When someone reports "the jack by the window is dead", you can go straight to the right port.

When it goes wrong

- **Desk dead after a move:** the outlet is fine but nobody patched its panel port to the switch.
- **[illegible]s on one desk:** the run is over length, or the cable was bent tighter [illegible]wn diameter.

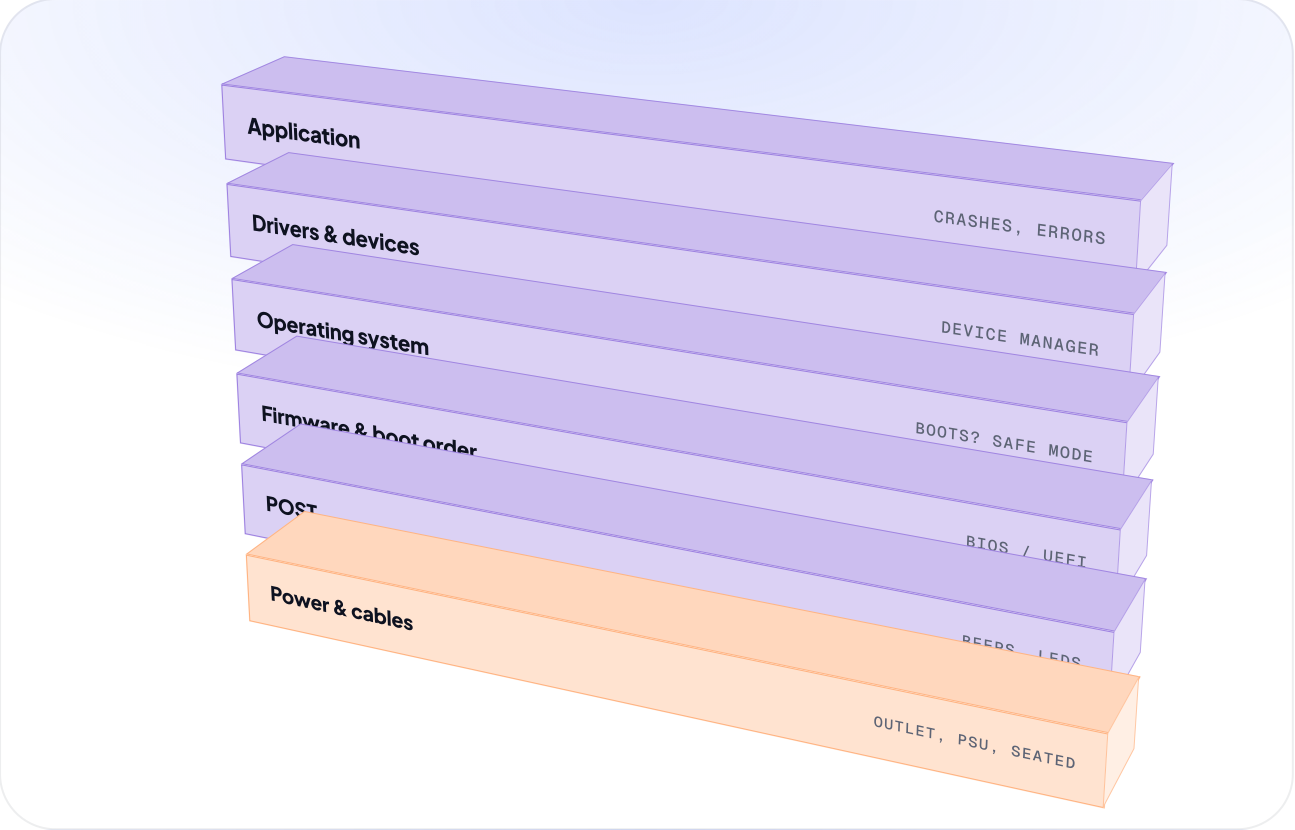
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- **Cable trays full of unlabelled cords:** label and trace now, before the next fault, not during it.

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Hardware Troubleshooting

A fixed order for a machine that will not start, will not boot or keeps failing.



Check from the bottom of the stack up: each layer needs the one beneath it working.

When a computer will not start, guessing wastes time. Work from the bottom up: each layer of a machine only works if the one beneath it does, so the first layer that fails is where the fault is.

1. Power and cables

- Is the outlet live? Try a lamp or a phone charger.
- Is the power cable seated at both ends, and the PSU switch on?
- On a laptop, remove the charger (and the battery if it comes out), hold the power button for 30 seconds, then try again with the charger only.

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The **Power-On Self-Test** checks the CPU, memory and graphics before anything loads.

- No fans, no lights: back to power.
- Fans spin but no display, often with **beep codes** or a flashing LED: usually memory or graphics. Reseat the RAM and try one stick at a time.
- Look up the beep pattern for that make; it names the part.

3. Firmware and boot order

- The machine reaches the logo but says "no boot device": check in BIOS/UEFI that the drive is detected and first in the boot order.
- A drive that is not detected at all is a cable, a port or a failed drive.

4. Operating system

- It starts to load, then restarts or shows a blue screen: note the stop code.
- Try **Safe Mode**. If it works there, the fault is a driver or a startup program, not hardware.

5. Drivers and devices

Open Device Manager and look for warning icons.

```
WINDOWS CMD

devmgmt.msc
```

- A device with a yellow triangle needs its driver reinstalled or rolled back.
- If the problem started after an update, roll that driver back first.

6. Application

Only once the machine itself is healthy is it worth looking at the program that fails: ...che, read its logs.

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TIP
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> Change one thing at a time and write down what you changed. If you swap the RAM and the cable together and it works, you still do not know which one was faulty.

Before you close the ticket

Tell the user what failed and what you did, in one sentence. If a part was replaced, note it against the asset so the next person sees the history.